

E.1 / PAPER 2 / QUESTION 3

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3. The diagram shows the three lowest energy levels of the hydrogen atom. The energy of each level is indicated.

diagram not to scale

-1.51 eV _____

-3.40 eV _____

-13.6 eV _____

- (a) State the number of transitions between these levels that result in the emission of photons.

[1]

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- (b) Calculate the largest photon wavelength for these transitions.

[3]

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